

The travelling salesman problem 5D

1 a $D_7 B_{12} C_8 E_{14} A_{19} D = 60$

b $E_8 C_{11} A_{13} B_7 D_{14} E = 53$
or

$E_8 C_{11} D_7 B_{13} A_{14} E = 53$

c The better upper bound is 53 since this is lower.

2 a $Z_{10} X_{15} Y_{40} V_{55} W_{30} S_{70} Z = 220$

b $X_{10} Z_{15} V_{40} Y_{45} W_{30} S_{55} X = 195$
 $V_{15} Z_{10} X_{15} Y_{45} W_{30} S_{75} V = 190$

c The better upper bound is 190 because it is lower.

3 The application to a single vertex has order n^2 , so application to entire network has order $n \times n^2 = n^3$

Hence the estimate for running time is $0.27 \times \left(\frac{20}{12}\right)^3 = 1.25s$

4 a $R_{150} S_{210} T_{120} U_{180} V_{300} W_{240} R = 1200$ minutes

b $U_{120} S_{150} R_{120} V_{150} T_{180} W_{270} U = 990$
and

$U_{120} T_{150} V_{120} R_{150} S_{240} W_{270} U = 1050$

c $V_{120} R_{150} S_{120} U_{120} T_{180} W_{300} V = 990$
and

$V_{120} R_{150} U_{120} S_{210} T_{180} W_{300} V = 1080$

and

$V_{120} R_{150} U_{120} T_{180} W_{240} S_{210} V = 1020$

d The better upper bound is 990 because it is lower.

- 5 a** Since $x < 126$, x is the smallest entry in column A, and the nearest neighbour route from A is ADEGCBFA, of total weight $639 + x$.

Since $x < 126$, x is the smallest remaining entry in column G after rows B, E, G and D have been deleted, so the nearest neighbour route from B is BEGDAFCB, of total weight $648 + x$.

$$\text{So } 1419 = 639 + x + 648 + x \Rightarrow 2x = 132 \Rightarrow x = 66$$

- b** 705 miles is an upper bound. In the previous part we have found upper bounds $648 + x = 714$ and $639 + x = 705$. The second one is better.